Define & Classify Slope Suitability

Land Suitability for Agrivoltaics as a Farmland Preservation Strategy in Rhode Island

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COURSE: Yale Environmental Data Science Capstone 2026

PURPOSE: Multi-Criteria Decision Analysis (MCDA) to identify optimal towns for solar-agriculture co-location.

DATA SOURCES: RIGIS (2020 Land Use, 2011 LiDAR, 1997 Municipality Lines)

SET UP & REPRODUCIBILITY

1. Load packages
2. setwd("/Users/brendanelba/Downloads/r_kay")
3. Define color palette for consistency across visuals

```
library(rmarkdown)
library(tidyr)
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.
0.0 ──
## ✔ dplyr     1.2.0     ✔ purrr     1.0.2
## ✔ forcats   1.0.0     ✔ readr     2.1.5
## ✔ ggplot2   4.0.2     ✔ stringr   1.5.1
## ✔ lubridate 1.9.4     ✔ tibble    3.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflict
s() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tidyterra)
```

```
## Warning: package 'tidyterra' was built under R version 4.4.3
```

```
##
## Attaching package: 'tidyterra'
##
## The following object is masked from 'package:stats':
##
##     filter
```

```
library(sf)
```

```
## Warning: package 'sf' was built under R version 4.4.3
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(terra)
```

```
## Warning: package 'terra' was built under R version 4.4.3
```

```
## terra 1.8.93
##
## Attaching package: 'terra'
##
## The following object is masked from 'package:tidyr':
##
##     extract
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:terra':
##
##     rescale
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(ggrepel) # Required for non-overlapping labels in bubble plots

green_palette <- colorRampPalette(c("#f7fcf9", "#00441b"))(4)
```

DOWNLOAD DATA

Data Sources:

1. ri_town_lines: RIGIS (1997) - Municipal boundaries
2. ri_land_cover: RIGIS (2020) - LULC Vector data
3. ri_elevation_dem: USGS (2011) - 1m LiDar Digital Elevation Models

All data projected to NAD83 / Rhode Island (ftUS) - EPSG:3438

```
setwd("/Users/brendanelba/Downloads/r_kay")

ri_town_lines <- st_read("ri_town_lines_1997/ri_town_lines_1997.shp")
```

```
## Reading layer `ri_town_lines_1997' from data source
##   `/Users/brendanelba/Downloads/r_kay/ri_town_lines_1997/ri_town_lines_199
7.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 396 features and 10 fields
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: 220310.4 ymin: 23048.49 xmax: 432040.9 ymax: 340916.6
## Projected CRS: NAD83 / Rhode Island (ftUS)
```

```
ri_land_cover <- st_read("land_cover_use_2020/land_cover_use_2020.shp") %>%
  filter(Descr_2020 %in% c("Idle Agriculture (abandoned fields and orchard
s)",
                           "Cropland (tillable)",
                           "Pasture (agricultural not suitable for tillage)",
                           "Orchards, Groves, Nurseries"))
```

```
## Reading layer `land_cover_use_2020' from data source
##   `/Users/brendanelba/Downloads/r_kay/land_cover_use_2020/land_cover_use_2
020.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 68618 features and 5 fields
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: 220000 ymin: 23051.56 xmax: 434591.9 ymax: 343511.1
## Projected CRS: NAD83 / Rhode Island (ftUS)
```

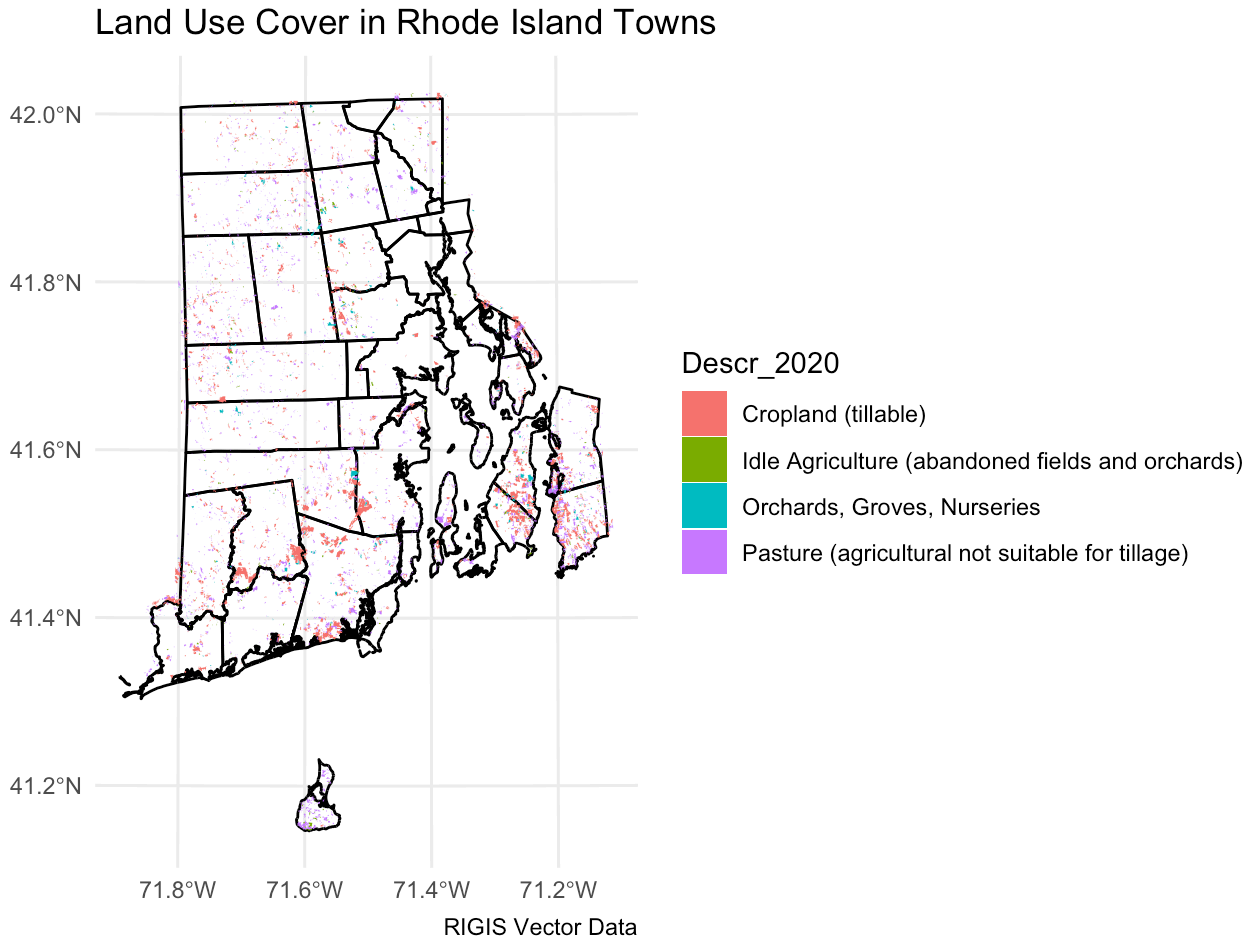
```
ri_elevation_dem <- list.files("ri_elevation_dem", pattern = "\\.img$", recur
sive = TRUE, full.names = TRUE)
```

DATA EXPLORATION & CLEANING

Viz 1: Rhode Island Land Use by Town

First look at the four agriculture land cover types of interest broken out by town and plotted on the Rhode Island state map.

```
(ri_land_use_by_town_map <- ggplot() +
    # Add municipality boundaries with black outlines
    geom_sf(data = ri_town_lines, fill = NA, color = "black", linewidth = 0.
5) +
    geom_sf(data = ri_land_cover, aes(fill = Descr_2020), color = NA) +
    theme_minimal() +
    labs(
      title = "Land Use Cover in Rhode Island Towns",
      caption = "RIGIS Vector Data"))
```



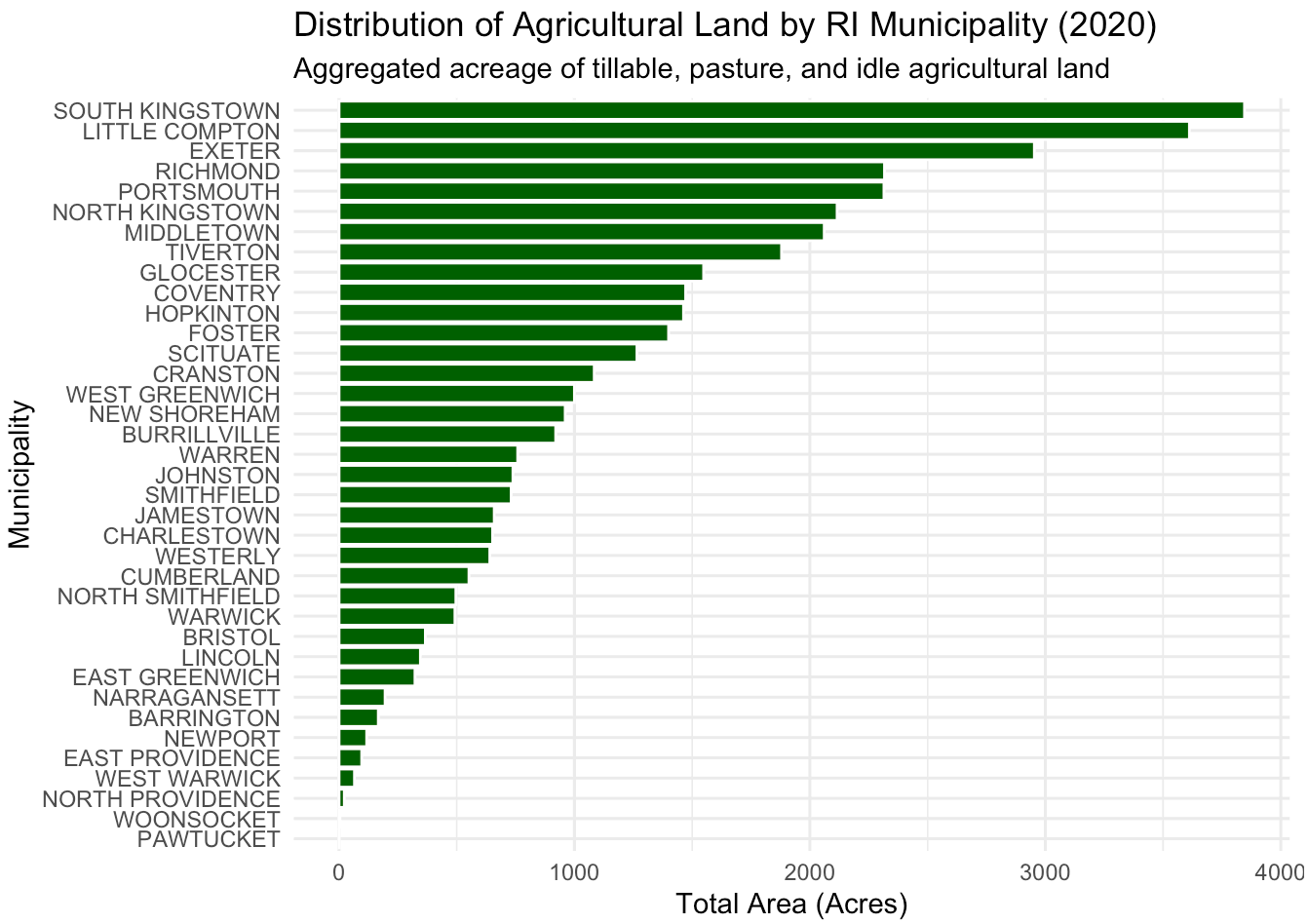

```
# Now identify which towns have the most agricultural land
  # First cut the land cover polygons by the municipality boundaries
intersections <- st_intersection(ri_land_cover, ri_town_lines)
```

```
## Warning: attribute variables are assumed to be spatially constant througho
ut
## all geometries
```

```
  # Now calculate the area of the new intersected pieces - farmland between t
own lines
intersections$area_sqm <- st_area(intersections)

# Create a summary plot
town_farm_land_use <- intersections %>%
  st_drop_geometry() %>% # Convert to a standard table for faster math
  group_by(NAME) %>%
  summarize(total_acres = sum(Acres_2020, na.rm = TRUE)) %>%
  arrange(desc(total_acres))
```

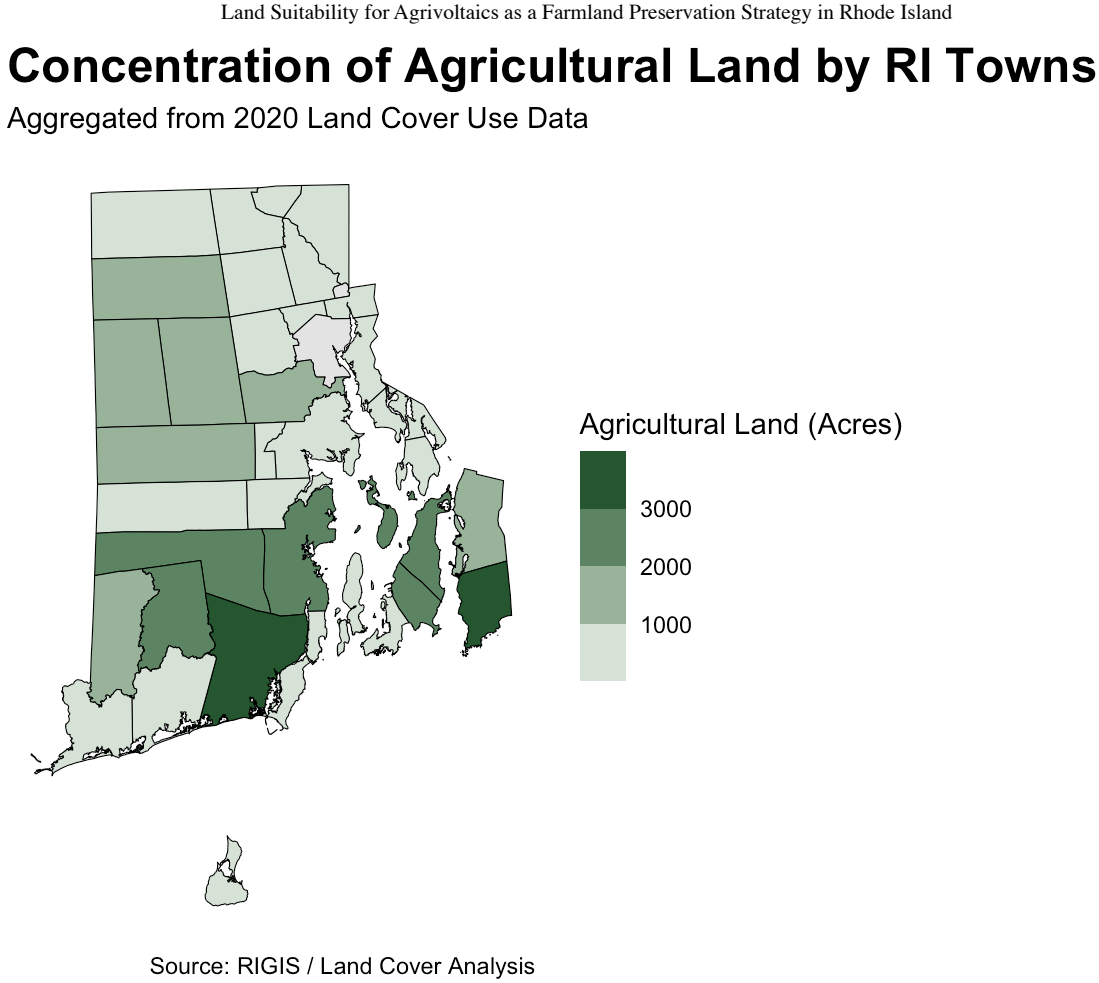
Viz 2: Farmland Acreage by Municipality

```
ggplot(town_farm_land_use, aes(x = reorder(NAME, total_acres), y = total_acre
s)) +
  geom_col(fill = "darkgreen", color = "white") +
  coord_flip() +  # Flip the chart to make municipality names readable
  labs(
    title = "Distribution of Agricultural Land by RI Municipality (2020)",
    subtitle = "Aggregated acreage of tillable, pasture, and idle agricultura
l land",
    x = "Municipality",
    y = "Total Area (Acres)"
  ) +
  theme_minimal()
```



Viz 3: Spatial Concentration of Ag Land in Rhode Island Towns

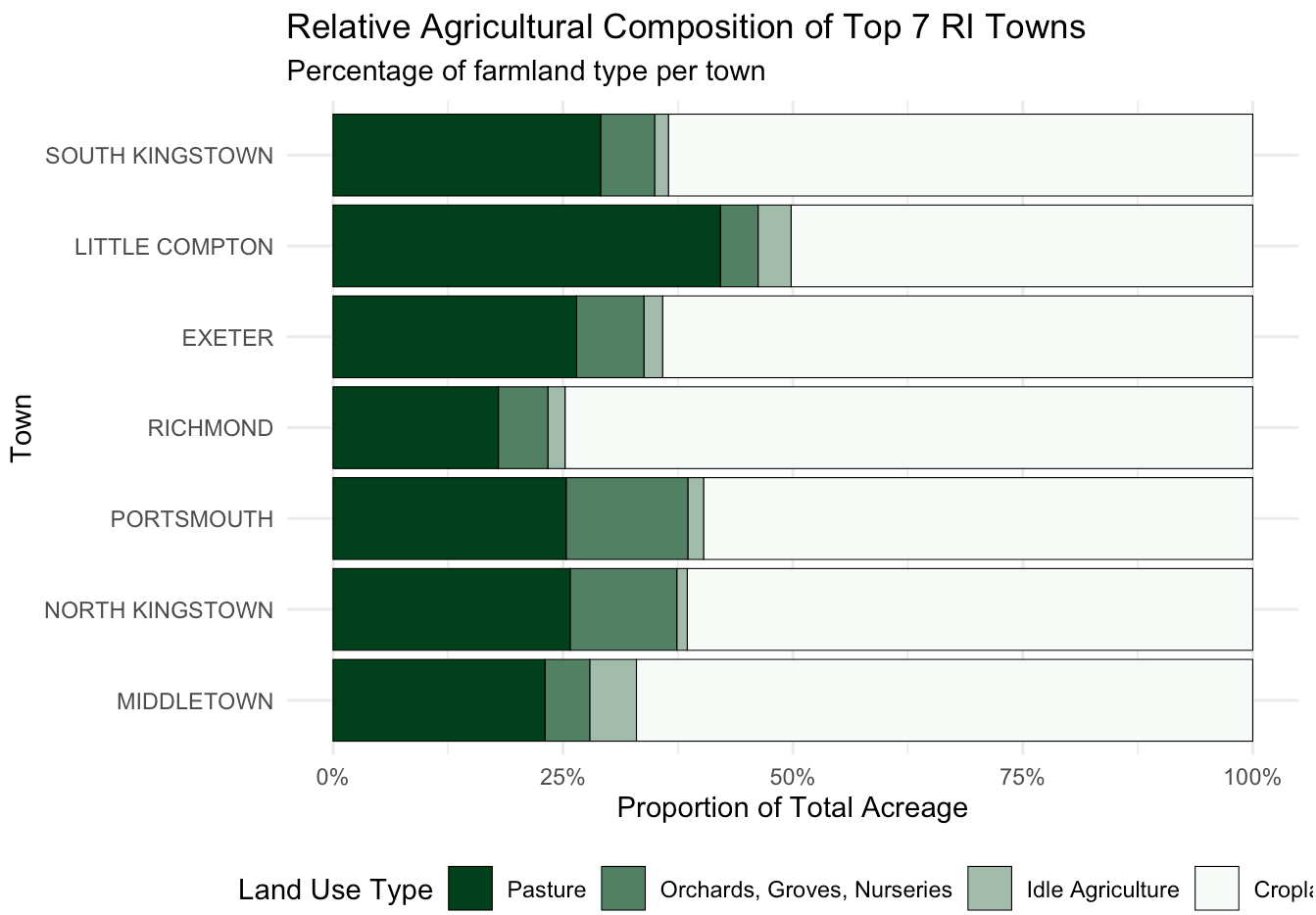
```
# Now join the summary back to the spatial town data
muni_map_data <- ri_town_lines %>%
  left_join(town_farm_land_use, by = "NAME") %>%
  st_transform(3438) #EPSG code - transforming from m^2 to US feet

## Viz 3: Map the spatial concentration of agricultural land in Rhode Island
Towns
ag_land_concentration <- ggplot(data = muni_map_data) +
  geom_sf(aes(fill = total_acres), color = "black", size = 0.2
)+
  scale_fill_stepsn(
    colors = c("#f7fcf5", "#00441b"),
    na.value = "grey90", # Color for missing data
    n.breaks = 5, # creating 5 bins
    name = "Agricultural Land (Acres)"
) +
  theme_minimal() +
  labs(title = "Concentration of Agricultural Land by RI Towns",
       subtitle = "Aggregated from 2020 Land Cover Use Data",
       caption = "Source: RIGIS / Land Cover Analysis"
) +
  theme(
    plot.title = element_text(size = 18, face = "bold"),
    axis.text = element_blank(),
    axis.title = element_blank(),
    panel.grid = element_blank()
)
print(ag_land_concentration)
```




```
## From viz 2 & 3, identify the 7 towns that have over 2000 total acres of fa
rmland
target_towns <- intersections %>%
    filter(NAME %in% c("SOUTH KINGSTOWN", "LITTLE COMPTON", "EXETER", "RICHMO
ND", "PORTSMOUTH", "NORTH KINGSTOWN", "MIDDLETOWN"))

# Agricultural Composition of Top 7 Towns
  # Filter the main data for these 7 towns and group by both Name and Type
muni_type_breakdown <- target_towns %>%
  st_drop_geometry() %>%
  group_by(NAME, Descr_2020) %>%
  summarize(Type_Acres = sum(Acres_2020, na.rm = TRUE), .groups = "drop")

# Change labels for clarity
my_labels <- c("Cropland (tillable)" = "Cropland",
               "Pasture (agricultural not suitable for tillage)" = "Pasture",
               "Orchards, Groves, Nurseries" = "Orchards, Groves, Nurseries",
               "Idle Agriculture (abandoned fields and orchards)" = "Idle Agr
iculture"
)
```

Viz 4: Percentage of Each Type of Farmland Cover

```
# A "normalized" stacked bar chart showing percentage of each type of farmlan
d cover
breakdown_bar_chart <- ggplot(muni_type_breakdown,
                              aes(x = reorder(NAME, Type_Acres, sum),
                                  y = Type_Acres,
                                  fill = Descr_2020)) +
  geom_col(position = "fill", color = "black", linewidth = 0.2) +
  coord_flip() +
  guides(fill = guide_legend(reverse = TRUE)) +
  scale_y_continuous(labels = scales::percent) + # Make axis to 0-100%
  scale_fill_manual(values = green_palette, name = "Land Use Type", labels =
my_labels) +
  labs(title = "Relative Agricultural Composition of Top 7 RI Towns",
       subtitle = "Percentage of farmland type per town",
       x = "Town", y = "Proportion of Total Acreage") +
  theme_minimal() +
  theme(legend.position = "bottom")

print(breakdown_bar_chart)
```



Elevation Data Pre-Upload Work

Breaking out what raster elevation data to download. Entire state elevation data is too large a file for this capstone

```
# To manage processing time, the project will breakout the 7 priority towns
target_towns <- intersections %>%
    filter(NAME %in% c("SOUTH KINGSTOWN", "LITTLE COMPTON", "EXETER", "RICHMO
ND", "PORTSMOUTH", "NORTH KINGSTOWN", "MIDDLETOWN"))

# Load the Tile Index ust downloaded
tiles <- st_read("tile_index/USGS_2011_LiDAR_index.shp")
```

```
## Reading layer `USGS_2011_LiDAR_index' from data source
##   `/Users/brendanelba/Downloads/r_kay/tile_index/USGS_2011_LiDAR_index.sh
p'
##   using driver `ESRI Shapefile'
## Simple feature collection with 434 features and 1 field
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: 210000 ymin: 20000 xmax: 440000 ymax: 350000
## Projected CRS: NAD83 / Rhode Island (ftUS)
```

```
# Find which tiles overlap with priority farmland
  # Creates a list of just the tiles actually needed
needed_tiles <- tiles[target_towns, ]

# See the list of IDs
print(needed_tiles$ID) # Or whichever column has the filename
```

```
##   [1] "0300001000" "0310001000" "0290001100" "0300001100" "0310001100"
##   [6] "0320001100" "0270001200" "0300001200" "0310001200" "0320001200"
##  [11] "0330001200" "0260001300" "0270001300" "0280001300" "0290001300"
##  [16] "0300001300" "0310001300" "0320001300" "0330001300" "0340001300"
##  [21] "0410001300" "0260001400" "0270001400" "0280001400" "0290001400"
##  [26] "0300001400" "0310001400" "0320001400" "0330001400" "0340001400"
##  [31] "0390001400" "0410001400" "0420001400" "0260001500" "0270001500"
##  [36] "0280001500" "0290001500" "0300001500" "0310001500" "0320001500"
##  [41] "0330001500" "0340001500" "0350001500" "0380001500" "0390001500"
##  [46] "0400001500" "0410001500" "0420001500" "0430001500" "0240001600"
##  [51] "0250001600" "0270001600" "0280001600" "0290001600" "0300001600"
##  [56] "0310001600" "0320001600" "0330001600" "0340001600" "0380001600"
##  [61] "0390001600" "0400001600" "0410001600" "0420001600" "0430001600"
##  [66] "0250001700" "0260001700" "0270001700" "0280001700" "0290001700"
##  [71] "0300001700" "0310001700" "0320001700" "0330001700" "0340001700"
##  [76] "0380001700" "0390001700" "0400001700" "0410001700" "0420001700"
##  [81] "0430001700" "0250001800" "0260001800" "0270001800" "0280001800"
##  [86] "0290001800" "0300001800" "0310001800" "0320001800" "0330001800"
##  [91] "0340001800" "0370001800" "0380001800" "0390001800" "0330001900"
##  [96] "0340001900" "0350001900" "0370001900" "0380001900" "0390001900"
## [101] "0340002000" "0350002000" "0360002000" "0370002000" "0380002000"
## [106] "0390002000"
```

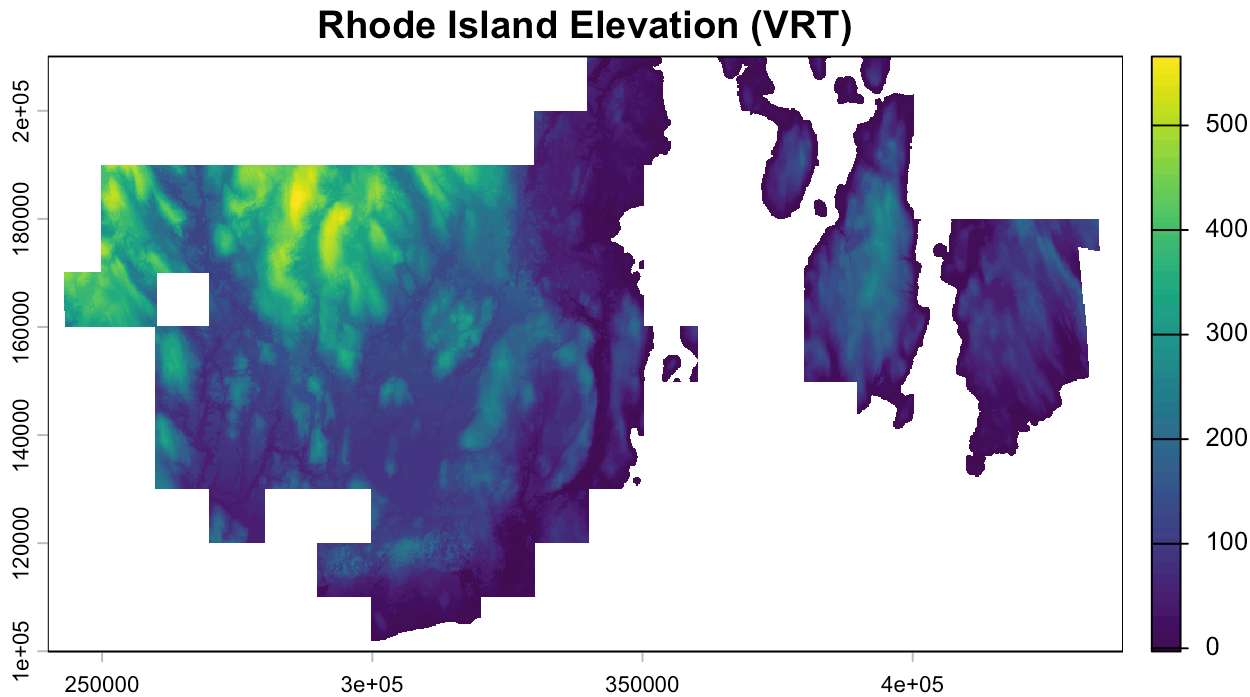
```
# Now that the IDs are known, the elevation raster data can be downloaded
  # Pull in the list of all the unzipped DEM files (see the "Download Data" s
ection)
```

Create Virtual Raster (VRT) Workflow

```
# Create a Virtual Raster for performance
  # This treats all 106 files as one single object called 'vrt_dem'
vrt_dem <- vrt(ri_elevation_dem, "ri_farmland_dem.vrt", overwrite = TRUE)

# Check it out
print(vrt_dem)
```

```
## class       : SpatRaster
## size        : 33560, 60643, 1  (nrow, ncol, nlyr)
## resolution  : 3.280833, 3.280833  (x, y)
## extent      : 239949.5, 438909.1, 99947.38, 210052.1  (xmin, xmax, ymin, y
max)
## coord. ref. : NAD83 / Rhode Island (ftUS)
## source      : ri_farmland_dem.vrt
## name        : ri_farmland_dem
## min value   :       -5.415652
## max value   :      585.934937
```

```
plot(vrt_dem, main = "Rhode Island Elevation (VRT)")
```

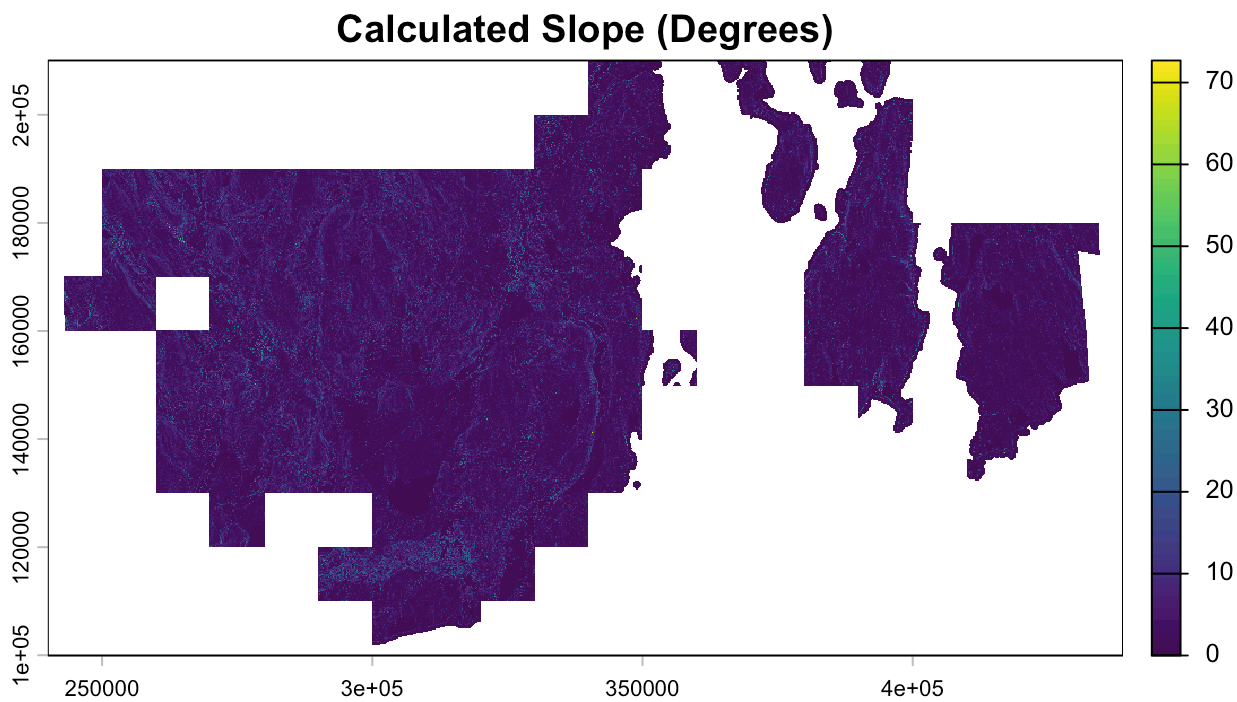


Calculate Slope

```
# Calculate slope in degrees
# 'filename' tells R to write the result to your disk instead of keeping it i
n RAM - because this is high-rez spatial data, keeping the results in the dis
k acts as a defense against the "out of memory" error
ri_slope <- terrain(vrt_dem, v = "slope", unit = "degrees",
                    filename = "ri_slope_output.tif",
                    overwrite = TRUE)
```

```
## |---------|---------|---------|---------|================================
========
```

```
plot(ri_slope, main = "Calculated Slope (Degrees)")
```



Define & Classify Slope Suitability

Slope (Degrees) | Suitability Score | Description

0 – 3° 4 Ideal / Flat

3 – 8° 3 Good

8 – 15° 2 Marginal (Higher Costs)

> 15° 1 Unsuitable / High Erosion

```
# Define the scoring matrix: [from, to, new_value]
m <- c(0, 3, 4,
       3, 8, 3,
       8, 15, 2,
       15, 90, 1)

reclass_matrix <- matrix(m, ncol=3, byrow=TRUE)

# Apply the scores
slope_scores <- classify(ri_slope, reclass_matrix, filename = "slope_suitabil
ity.tif", overwrite = TRUE)
```

```
## |---------|---------|---------|---------|================================
========
```

Convert Land Cover to SpatRaster

```
# Convert the intersections to a terra SpatVector
  # Filter for the top 7 Towns
farmland_vect <- vect(target_towns)

# Combine all small farm polygons into one shape per town - only want to look
at the seven towns, not the individual plots of farmland
farmland_town_clean <- aggregate(farmland_vect, by = "NAME")

# Force-project the vector to match the raster exactly
farmland_vect_projected <- project(farmland_town_clean, crs(slope_scores))
```

Mask by Ag Land

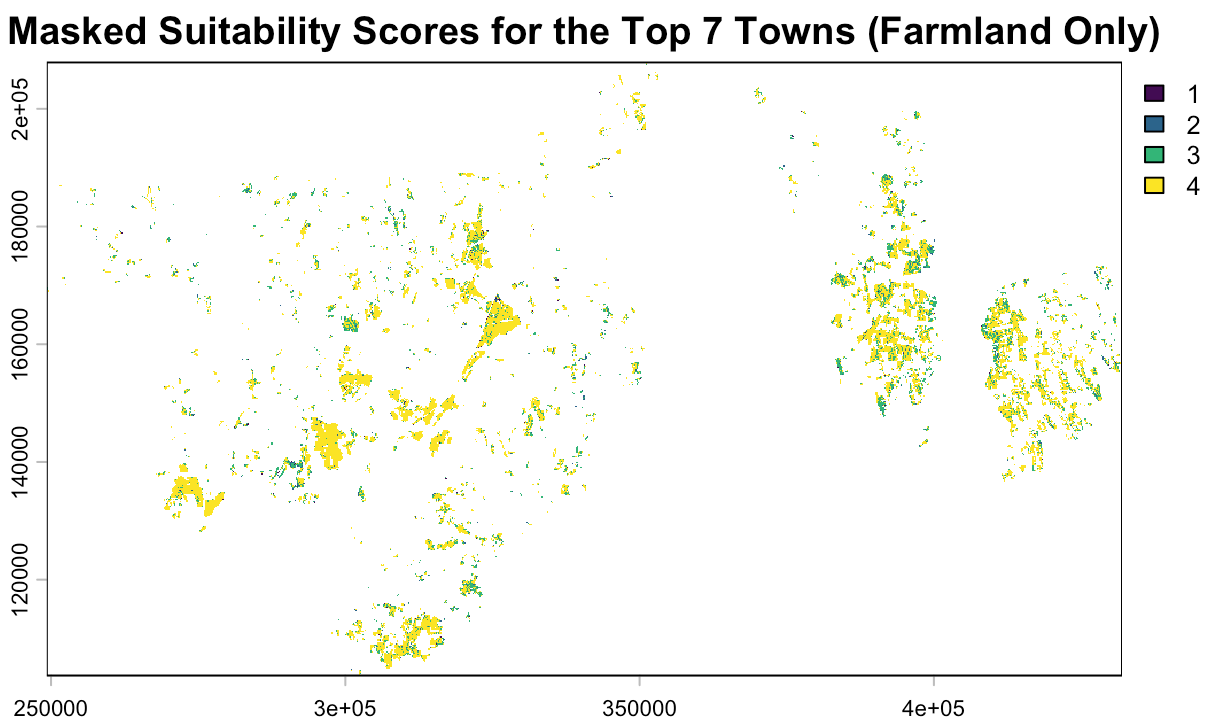
```
# Keeping the raster values only where the polygons are
  # (aka masking slope data to agricultural boundaries only)
slope_cropped <- crop(slope_scores, farmland_vect_projected)
```

```
## |---------|---------|---------|---------|================================
========
```

```
farmland_slope_masked <- mask(slope_cropped, farmland_vect_projected)
```

```
## |---------|---------|---------|---------|================================
========                                    |---------|---------|------
---|---------|========================================
```

```
# Plot the farmland
plot(farmland_slope_masked, main = "Masked Suitability Scores for the Top 7 T
owns (Farmland Only)")
```




```
# Calculating the mean suitability score for each town
town_results <- zonal(farmland_slope_masked, farmland_vect_projected["NAME"],
fun = "mean", na.rm = TRUE)
```

```
## |---------|---------|---------|---------|==================================
========                                    |---------|---------|------
---|---------|=======================================
|---------|---------|---------|---------|=====================================
=====                                       |---------|---------|---------
|---------|=======================================
|---------|---------|---------|---------|=====================================
=====                                       |---------|---------|---------
|---------|=======================================
|---------|---------|---------|---------|=====================================
=====
```

```
# CHECK: If R dropped the names, then manually add them back
if (ncol(town_results) == 1) {
  # Create a clean data frame using the names from our aggregated vector
  town_results <- data.frame(
    NAME = farmland_vect_projected$NAME,
    slope_score = town_results[,1]
  )
} else {
  # If R kept the names, just rename the columns
colnames(town_results) <- c("NAME", "slope_score")
}

# Sort so the best town is at the top
town_results <- town_results %>%
  arrange(desc(slope_score))

print(town_results)
```

```
##              NAME slope_score
## 1        RICHMOND    3.742477
## 2 SOUTH KINGSTOWN    3.732623
## 3 NORTH KINGSTOWN    3.700816
## 4  LITTLE COMPTON    3.687490
## 5      MIDDLETOWN    3.659841
## 6          EXETER    3.631879
## 7      PORTSMOUTH    3.595101
```

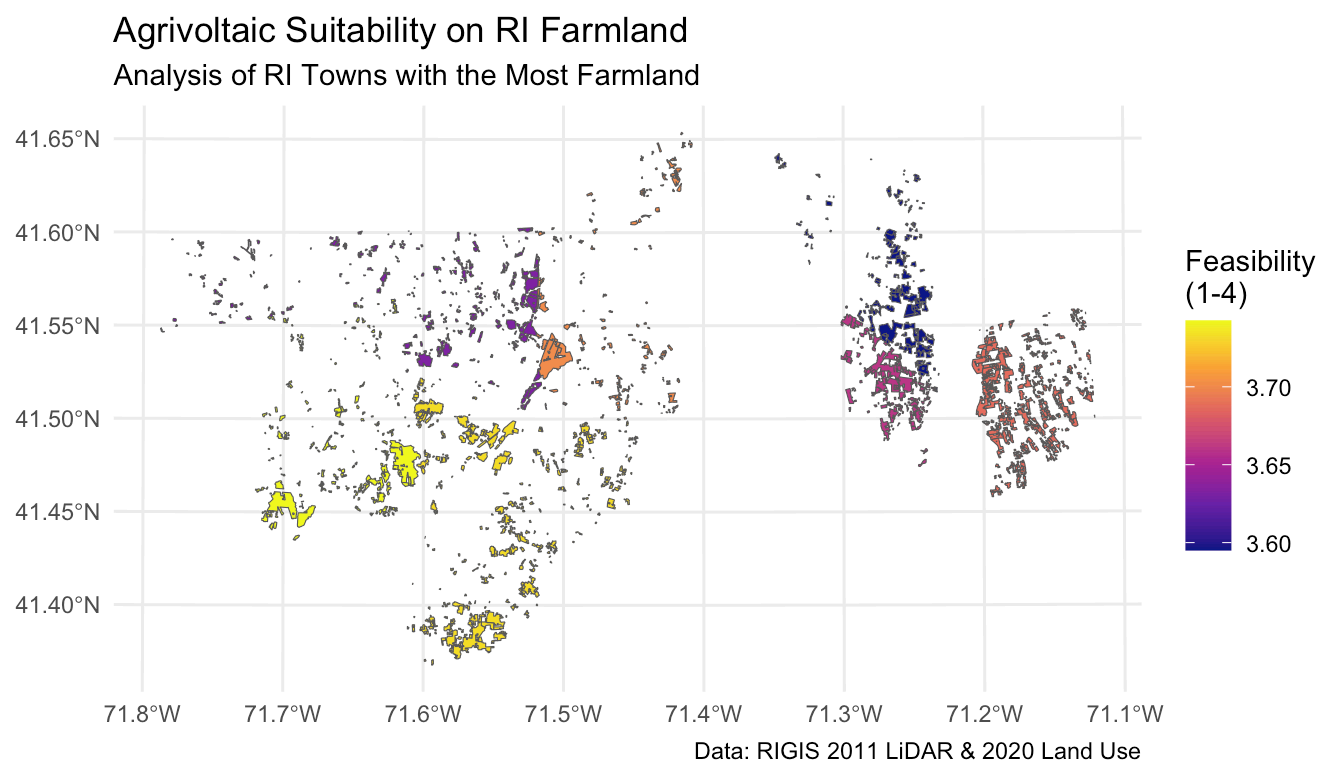
MULTI-CRITERIA ANALYSIS

We are looking at existing agriculture land that could have Agrivoltaics implemented. The goal is to find the cheapest, easiest land to build on without upsetting active farmers or changing existing crops.

###Calculate Mean Slope Suitability

CRITERIA 1: Slope Suitability (35%)

```
# Calculate Mean Slope Suitability per Town
# Join the math back to the shapes for a final map
final_suitability_map <- merge(target_towns, town_results, by.x = "NAME", by.
y = "NAME")

# Viz 5: Plot Town Slope Suitability for Agrivoltaics
slope_suitability_plot <- ggplot(final_suitability_map) +
  geom_sf(aes(fill = slope_score)) +
  scale_fill_viridis_c(option = "plasma", name = "Feasibility\n(1-4)") +
  labs(title = "Agrivoltaic Suitability on RI Farmland",
       subtitle = "Analysis of RI Towns with the Most Farmland",
       caption = "Data: RIGIS 2011 LiDAR & 2020 Land Use") +
  theme_minimal()

print(slope_suitability_plot)
```



###Scoring Land Used for Agriculture

CRITERIA 2: Land Use Type (20% Weight)

- *Idle Agriculture (Score 4): Best. No active crops to displace; often clear of large trees.*
- *Pasture (Score 3): Great. Flat, cleared, and low-impact to the soil to install racking.Cropland (Score 2): Marginal. High value for food; installation disrupts active tillable soil.*

- *Orchards/Nurseries (Score 1): Worst. Removing trees is expensive and significantly changes the land use.*

```
# Create the scoring "lookup" table
land_use_scores <- data.frame(
  Descr_2020 = c(
    "Idle Agriculture (abandoned fields and orchards)", # Highest: Low confli
ct
    "Cropland (tillable)",                              # Moderate: High soil
value and tillable
    "Pasture (agricultural not suitable for tillage)",  # High: Perfect for s
heep/grazing
    "Orchards, Groves, Nurseries"),                     # Lowest: High cleari
ng costs/permanent crops
  cover_score = c(4, 3, 2, 1)
)

# Joining the scores to the 'intersections' data
land_use_scores_jnd <- intersections %>%
  left_join(land_use_scores, by = "Descr_2020")%>%
  filter(NAME %in% c("SOUTH KINGSTOWN", "LITTLE COMPTON", "EXETER", "RICHMON
D", "PORTSMOUTH", "NORTH KINGSTOWN", "MIDDLETOWN"))

# Calculate the area-weighted mean of agricultural land use for each town
avg_land_use_score <- land_use_scores_jnd %>%
  st_drop_geometry() %>%
  group_by(NAME) %>%
  # Multiply score by acreage, then divide by total town acreage
  summarize(avg_land_use_score = sum(cover_score * Acres_2020) / sum(Acres_20
20)) %>%
  arrange(desc(avg_land_use_score))

print(avg_land_use_score)
```

```
## # A tibble: 7 × 2
##   NAME            avg_land_use_score
##   <chr>                        <dbl>
## 1 RICHMOND                      2.73
## 2 MIDDLETOWN                    2.72
## 3 EXETER                        2.61
## 4 SOUTH KINGSTOWN               2.61
## 5 LITTLE COMPTON                2.53
## 6 NORTH KINGSTOWN               2.52
## 7 PORTSMOUTH                    2.50
```

Total Available Farmland Score

CRITERIA 3: Plot Efficiency & Capacity (45% Weight)

Thresholds based on industry standards for Utility vs. Community Scale solar

####1. Utility Ratio (60%): Reward plots over >25 acres size

```
1. Efficiency Score (60%): Reward the average size.
    # > 70 ratio = 4 points (Utility Scale)
    # 66-69 ratio = 3 points (Community Scale)
    # 62-65 ratio = 2 points (Standard)
    # >61 ratio = 1 points (Marginal)
```

2. Capacity Score (40%): Reward the total volume.

```
    # > 3,500 acres = 4 points
    # 3,499–2,500 acres = 3 points
    # 2,199–2,499 acres = 2 points
    # < 2,100 acres = 1 point
```

```
# Utility Scale Analysis: Plots > 25 Acres

town_viability_ranks <- intersections %>%
  filter(Acres_2020 >= 2.0) %>%
  filter(NAME %in% c("SOUTH KINGSTOWN", "LITTLE COMPTON", "EXETER",
                     "RICHMOND", "PORTSMOUTH", "NORTH KINGSTOWN", "MIDDLETOW
N")) %>%
  st_drop_geometry() %>%
  group_by(NAME) %>%
  summarize(
    total_viable_acreage = sum(Acres_2020, na.rm = TRUE),
    avg_viable_plot_size = mean(Acres_2020, na.rm = TRUE),

    # Calculate how much acreage comes from plots > 25 acres
    utility_scale_acreage = sum(Acres_2020[Acres_2020 > 25], na.rm = TRUE),

    # Create the Utility Ratio (AKA what % of town ag land is > 25 acre plot
s?)
    utility_ratio = (utility_scale_acreage / total_viable_acreage) * 100
  ) %>%
  mutate(
    # Efficiency Score taken from Utility Ratio
    # This should reward towns where the majority of land is in large, effici
ent blocks
    size_score = case_when(
      utility_ratio >= 70 ~ 4, # Over 70% the town's farmland is Utility-Scal
e
      utility_ratio >= 66 ~ 3,
      utility_ratio >= 62 ~ 2,
      TRUE                ~ 1
    ),

    # 2. Capacity Score (Total Volume)
    capacity_score = case_when(
      total_viable_acreage >= 3500 ~ 4,
      total_viable_acreage >= 2500 ~ 3,
      total_viable_acreage >= 2100  ~ 2,
      TRUE                         ~ 1
    ),

    # Finally, the Combined Viability (60% Ratio Efficiency / 40% Total Capac
ity)
    combined_viability_score = (size_score * 0.6) + (capacity_score * 0.4)
  ) %>%
  arrange(desc(combined_viability_score))

print(town_viability_ranks)
```

```
## # A tibble: 7 × 8
##   NAME           total_viable_acreage avg_viable_plot_size utility_scale_a
creage
##   <chr>                         <dbl>                <dbl>
<dbl>
## 1 RICHMOND                      2209.                 17.1
1551.
## 2 LITTLE COMPTON                3503.                 15.4
2187.
## 3 SOUTH KINGSTO…                3726.                 15.7
2272.
## 4 MIDDLETOWN                    2022.                 22.2
1402.
## 5 NORTH KINGSTO…                2029.                 19.1
1382.
## 6 PORTSMOUTH                    2273.                 18.0
1482.
## 7 EXETER                        2812.                 14.5
1739.
## # ℹ 4 more variables: utility_ratio <dbl>, size_score <dbl>,
## #   capacity_score <dbl>, combined_viability_score <dbl>
```

FINAL WEIGHTED OVERLAY ANALYSIS

```
If this analysis were to be implemented on a larger research scale, instead o
f a weighted overlay analysis, it would be recommended to implement an Analyt
ic Hierarchy Process (AHP) which would move the analysis from the current "be
st guess" weights to a mathematically validated Priority Vector.

Model Logic: Land Viability (45%) + Slope (35%) + Land Cover (20%)
```

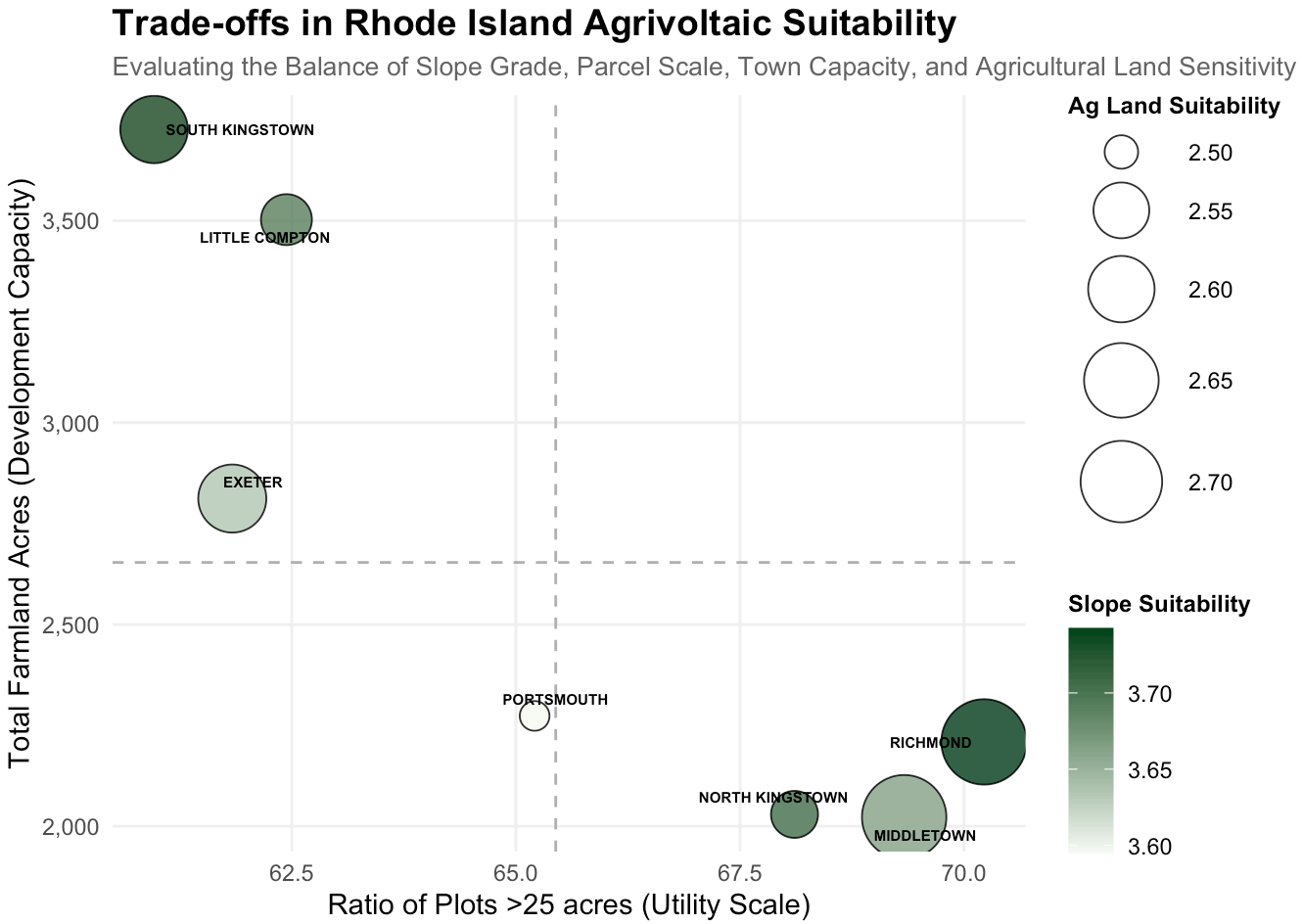
```
# Joining it all together
final_capstone_results <- town_results %>%
  inner_join(avg_land_use_score, by = "NAME") %>%
  inner_join(town_viability_ranks, by = "NAME") %>%

  # The final weighted index
  mutate(Total_Suitability = (slope_score * 0.35) +
                             (combined_viability_score * 0.45) +
                             (avg_land_use_score * 0.2)) %>%
  arrange(desc(Total_Suitability))

# Assign Final Rank
final_capstone_results$final_rank <- 1:nrow(final_capstone_results)

# Print the results to verify
print(final_capstone_results %>%
        select(final_rank, NAME, Total_Suitability, slope_score, combined_via
bility_score, avg_land_use_score))
```

```
##   final_rank           NAME Total_Suitability slope_score
## 1          1        RICHMOND          3.296120    3.742477
## 2          2  LITTLE COMPTON          3.057178    3.687490
## 3          3 SOUTH KINGSTOWN          2.817732    3.732623
## 4          4      MIDDLETOWN          2.815542    3.659841
## 5          5 NORTH KINGSTOWN          2.789579    3.700816
## 6          6      PORTSMOUTH          2.658070    3.595101
## 7          7          EXETER          2.602998    3.631879
##   combined_viability_score avg_land_use_score
## 1                      3.2           2.731265
## 2                      2.8           2.532782
## 3                      2.2           2.606570
## 4                      2.2           2.722989
## 5                      2.2           2.521469
## 6                      2.0           2.498925
## 7                      1.8           2.609200
```

VISUALIZING THE RESULTS

VIZ A: Efficiency vs. Capacity Bubble Plot

```
# Create a bubble plot to show the technical constraints

# Calculate averages for the quadrant cross hairs
ggplot(final_capstone_results, aes(x = utility_ratio, y = total_viable_acreag
e)) +
  # Crosshairs based on the mean of raw values
  geom_vline(xintercept = mean(final_capstone_results$utility_ratio, na.rm =
TRUE),
             linetype = "dashed", color = "grey70") +
  geom_hline(yintercept = mean(final_capstone_results$total_viable_acreage, n
a.rm = TRUE),
             linetype = "dashed", color = "grey70") +

  # Point size is tied to the combined viability (Efficiency + Capacity)
  geom_point(aes(size = avg_land_use_score, fill = slope_score),
             shape = 21, color = "black", alpha = 0.85) +

  geom_text_repel(aes(label = NAME), fontface = "bold", size = 2) +

  scale_size_continuous(range = c(5, 15), name = "Ag Land Suitability") + # B
ubble Size: Slope suitability (How flat is it?).
  scale_fill_gradient(low = "#f7fcf5", high = "#00441b", name = "Slope Suitab
ility") +
  scale_y_continuous(labels = scales::comma) +

  labs(title = "Trade-offs in Rhode Island Agrivoltaic Suitability",
       subtitle = "Evaluating the Balance of Slope Grade, Parcel Scale, Town
Capacity, and Agricultural Land Sensitivity",
       x = "Ratio of Plots >25 acres (Utility Scale)", # X-Axis: Efficiency
(Are the plots big enough to make construction costs worth it?).
       y = "Total Farmland Acres (Development Capacity)") + # Y-Axis: Total C
apacity (How much solar can we build?).
  theme_minimal() +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "grey95"),
    legend.title = element_text(size = 9, face = "bold"),
    legend.background = element_rect(fill = "white", color = NA),
    plot.title = element_text(size = 14, face = "bold"),
    plot.subtitle = element_text(size = 10, color = "grey40")
    )
```


Trade-offs in Rhode Island Agrivoltaic Suitability
Evaluating the Balance of Slope Grade, Parcel Scale, Town Capacity, and Agricultural Land Sensitivity
SOUTH KINGSTOWN
LITTLE COMPTON
EXETER
PORTSMOUTH
RICHMOND
NORTH KINGSTOWN
MIDDLETOWN
3,500
3,000
2,500
2,000
Total Farmland Acres (Development Capacity)
62.5
65.0
67.5
70.0
Ratio of Plots >25 acres (Utility Scale)
Ag Land Suitability
2.50
2.55
2.60
2.65
2.70
Slope Suitability
3.70
3.65
3.60

This visualization segments the Top 7 towns into quadrants based on their Average Plot Size (X-axis) and Total Farmland Acres (Y-axis). Middletown occupies the far-right, while South Kingstown holds the top-left. Bubble Size (Slope Suitability): A larger bubble indicates flatter terrain. Bubble Color (Farmland Cover): Darker bubbles are towns with greater idle agriculture and cropland, while lighter bubbles indicate greater concentrations of pasture and orchards.

Top-Right (High Efficiency, High Capacity): These would be "Premier" zones. A town here would have both a high volume of agricultural land and large individual parcels. This could allow for utility-scale solar arrays that achieve economies of scale during construction.

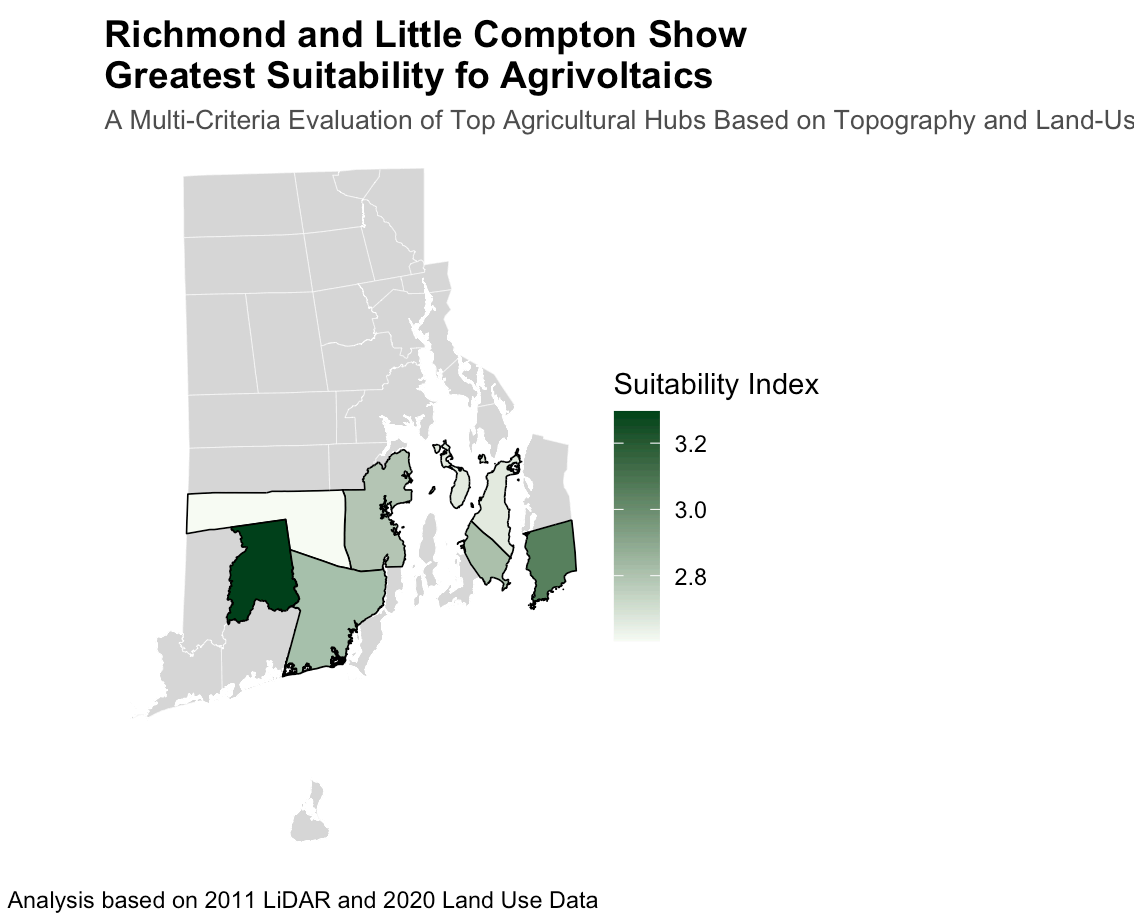
Top-Left (High Capacity, Low Efficiency): These towns have a larger amount of farmland, but it is fragmented into smaller plots. Development here would likely require many smaller solar sites rather than one large project.

Bottom-Right (High Efficiency, Low Capacity): These towns have larger plots but limited total acreage. These are ideal achieving economies of scale.

Interpretation: Middletown, Richmond, and South Kingstown emerge as Rhode Island’s primary agrivoltaic hubs due to a rare convergence of flat topography and expansive agricultural acreage. While South Kingstown serves as a capacity powerhouse offering the state's largest total volume of farmland for community-scale solar, Middletown and Richmond lead in operational efficiency. These two towns boast a higher concentration of large, contiguous, utility-scale plots that significantly lower per-megawatt installation costs for developers. This bubble plot illustrates the technical trade-offs that need to be considered in agrivoltaic site selection.

VIZ B: "Final Suitability" Choropleth Map

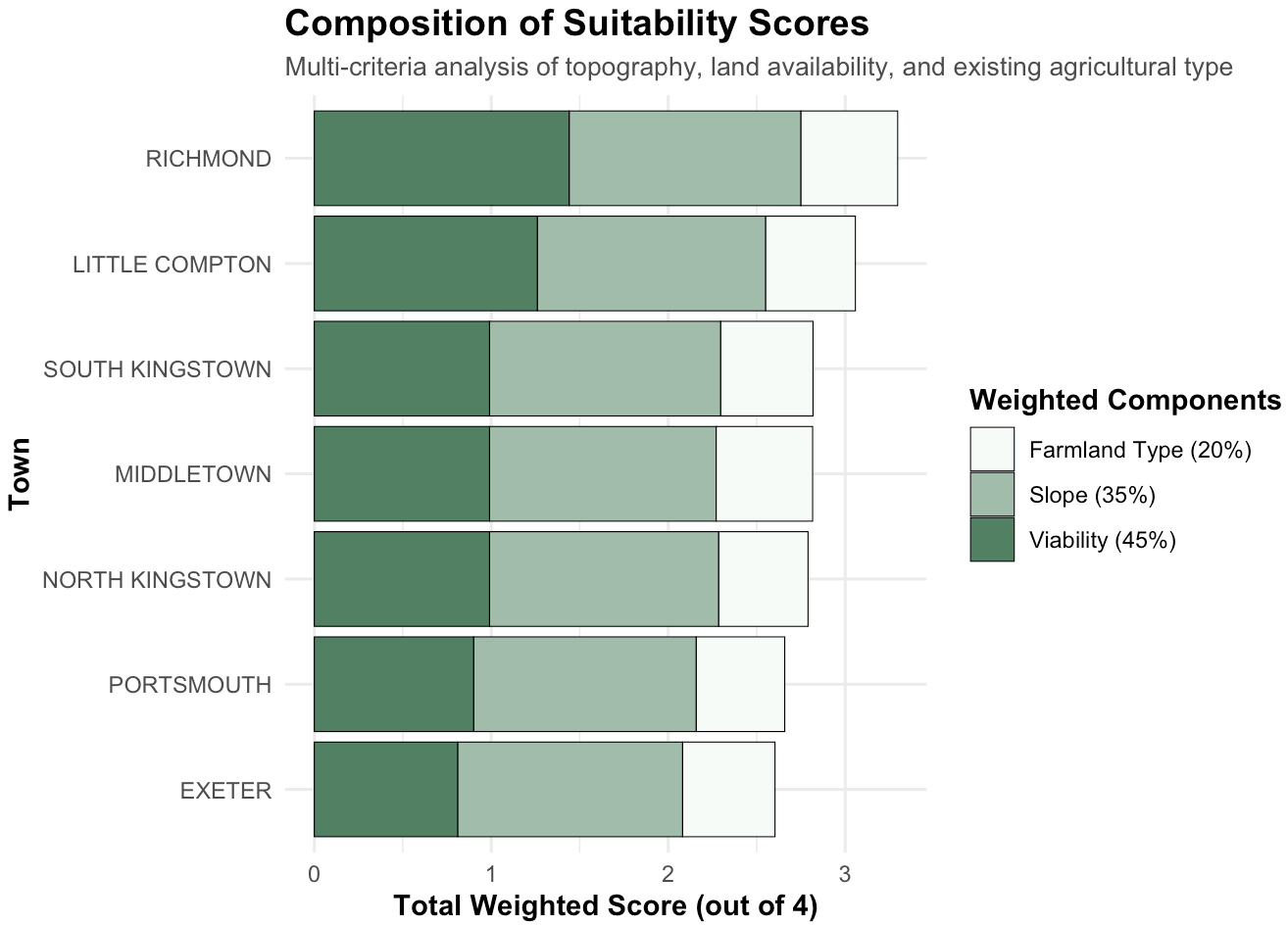
```
# Join results back to spatial data
final_map_data <- ri_town_lines %>%
  inner_join(final_capstone_results, by = "NAME")

write_csv(final_map_data, "final_suitability_data")

ggplot() +
  geom_sf(data = ri_town_lines, fill = "grey85", color = "white", linewidth =
0.1) + # base map

  geom_sf(data = final_map_data, aes(fill = Total_Suitability), color = "blac
k", linewidth = 0.3) + # suitability data on top 7 towns

  scale_fill_gradient(low = "#f7fcf5", high = "#00441b",
                      name = "Suitability Index",
                      na.value = "grey95") +

  labs(title = "Richmond and Little Compton Show \nGreatest Suitability fo Ag
rivoltaics",
       subtitle = "A Multi-Criteria Evaluation of Top Agricultural Hubs Based
on Topography and Land-Use",
       caption = "Analysis based on 2011 LiDAR and 2020 Land Use Data") +
  theme_void() +
  theme(legend.position = "right",
        plot.title = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(size = 10, color = "grey30"),
        plot.margin = margin(10, 10, 10, 10)
  )
```




```
The map displays the spatial concentration of the final Suitability Index acr
oss the study area (Top 7 Priority Towns), with the darkest green shades indi
cating the highest-ranking municipalities.

Interpretation: Richmond and Little Compton stand out as having the greatest
suitability for implementing agrivoltaics. Therefore, these towns should be t
he priority focus for policy makers, solar farm developers and town planners.
These two towns should be examined further to develop land ordinances that su
pport the dual-location of solar energy and agricultural properties. Preservi
ng the land as Farm, Forest, and Open Space for farmland preservation.
```

VIZ C: Scoring Composition Stacked Bar

```
# Stacked bar chart showing how much each town's score was influenced by each
factor (slope grade, plot size and total acreage of ag land per town, and lan
d cover type). This is a way to defend the weighting choices.

# Reshape data for plotting
plot_breakdown <- final_capstone_results %>%
  mutate(Slope_Component = slope_score * 0.35,
         Viability_Component = combined_viability_score * 0.45,
         Cover_Component = avg_land_use_score * 0.2) %>%
  select(NAME, Slope_Component, Viability_Component, Cover_Component) %>%
  pivot_longer(cols = -NAME, names_to = "Metric", values_to = "Score")

ggplot(plot_breakdown, aes(x = reorder(NAME, Score, sum), y = Score, fill = M
etric)) +
  geom_col(color = "black", linewidth = 0.2) +
  coord_flip() +
  scale_fill_manual(values = green_palette,
                    labels = c("Farmland Type (20%)", "Slope (35%)", "Viabili
ty (45%)"),
                    name = "Weighted Components") +
  labs(title = "Composition of Suitability Scores",
       subtitle = "Multi-criteria analysis of topography, land availability,
and existing agricultural type",
       x = "Town", y = "Total Weighted Score (out of 4)") +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "grey30"),
    axis.title = element_text(face = "bold"),
    legend.title = element_text(face = "bold")
)
```




```
This chart breaks down the final score for each town into its three weighted
components: Viability (45%), Slope (35%), and Land Cover (20%).

Interpretation: This stacked bar chart provides a transparent breakdown of th
e Multi-Criteria Decision Analysis (MCDA), illustrating how each of the three
weighted factors contributes to a town's final suitability score. While Richm
ond leads in total weighted score, the chart reveals a high degree of balance
across the top five municipalities. The narrow margin between the top contend
ers suggests that Rhode Island possesses a resilient "cluster" of potential a
grivoltaic hubs in both South County and the East Bay, rather than one single
outlier.
```

END CODE

```
```